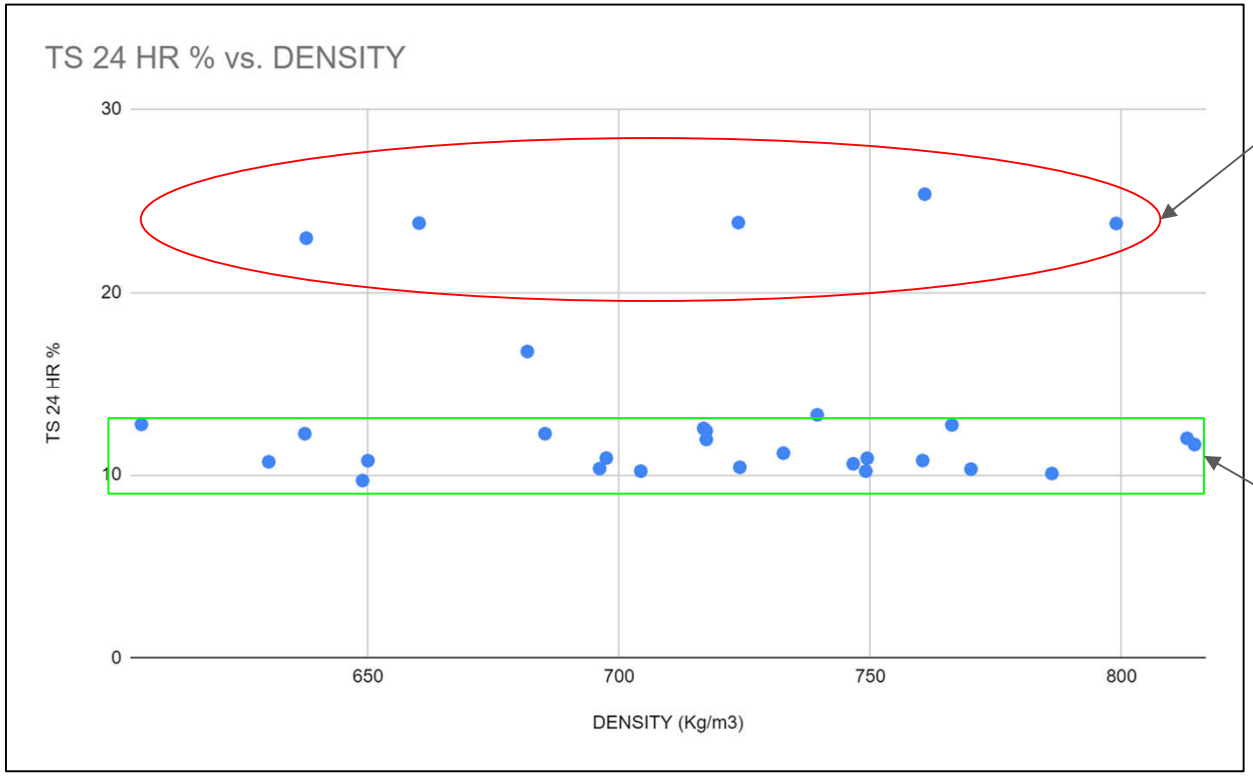
Simplified Comparison Table

	Oply 12mm		Plywood 12mm		Remark
	min	max	min	max	
Density (kg/m3)	600	725	500	600	**Plywood is lighter** by approx. 20% ~ 25%.
Modulus of Rupture (N/mm2)	30	50	30	60	Plywood has two axis - one axis is always greater than the other. **Oply MOR is within the range.** no different in two axis.
Thickness Swelling 24 hrs	8%	12%	8%	14%	Both are about **the same**.
Internal Bonding (N/mm2)	1	2	Not Applicable		To measure the strength of core. Note: for plywood - Chisel Test for the bonding quality between plies. IB of **Oply is better than typical particle board & MDF** (at range of 0.8)
Surface Soundness	Not yet carried out		Not Applicable		To measure the quality of surface.
Face Screw Withdrawal (N)	1,900	2,200	1,100	1,400	**Oply is better** than plywood in both face and edge.
Edge Screw Withdrawal (N)	1,400	1,900	1,050	1,300	

Oply Test Result - Analysis 1 - Thickness Swelling 24 hr vs. density

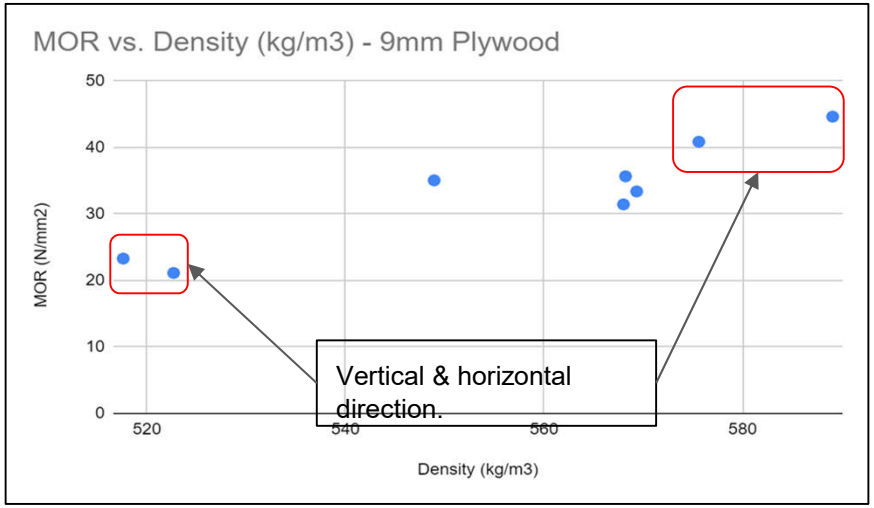


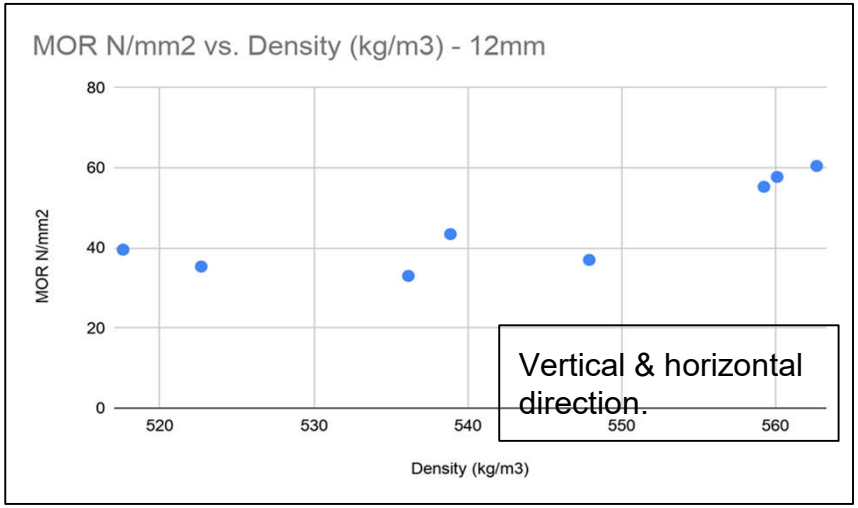
Unsatisfied thickness swelling is due to the curing temperature. (160 Dgr Celsus)

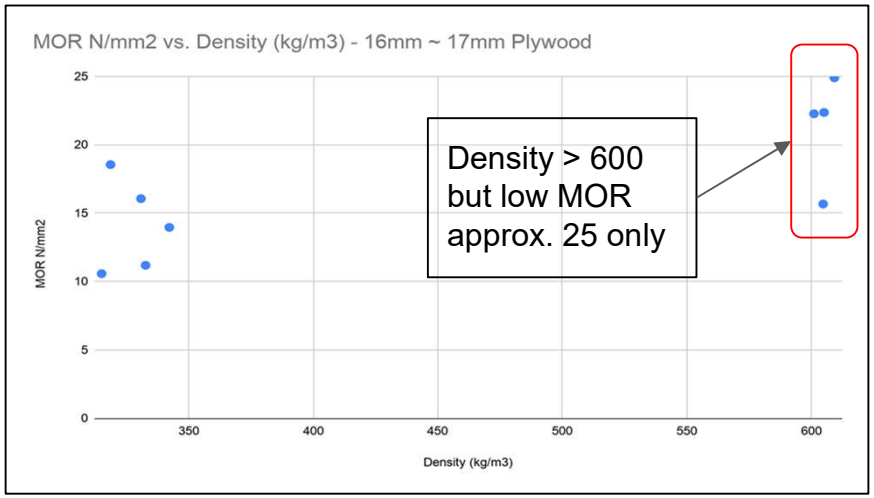
Good thickness swelling result shows both in *1.14cm, *3.46cm & *4.32cm fiber length.

*average length.

Doesn't show correlation between thickness swelling with density.







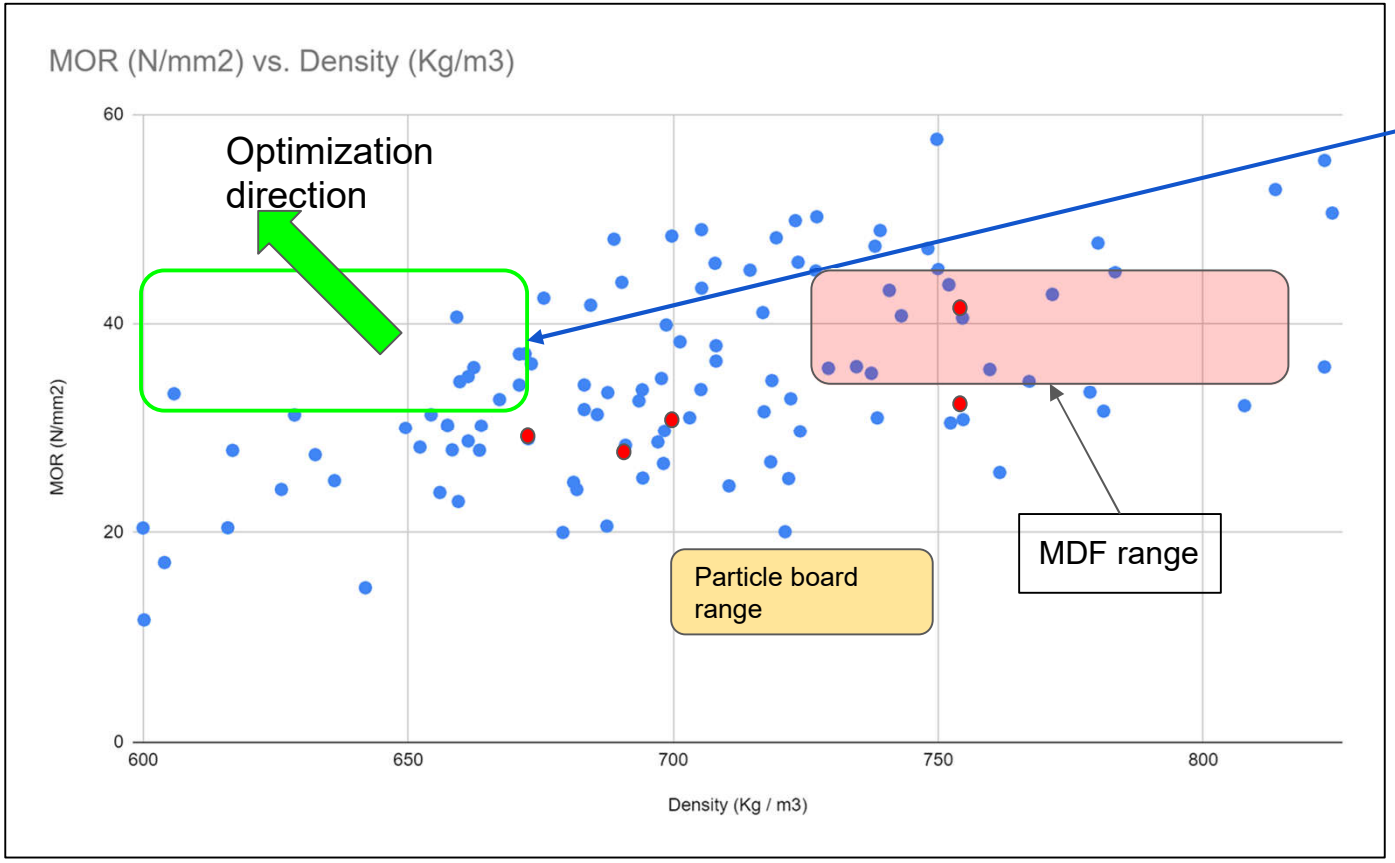
PLYWOOD MOR vs. Density Summary:

1.The density and MOR are proportional.

2. Same density NOT necessary shows similar MOR. Comparing 16mm & 12mm, approx. same density, 16mm MOR is much lower than 12mm MOR even 16mm is thicker than 12mm, 16mm max MOR is only 25N/mm2 whereas 12mm min MOR is above 30N/mm2.

3. Different orientation shows different MOR by approx double in value.

Oply Test Result - Analysis 2 - MOR vs. density



Commonality:-

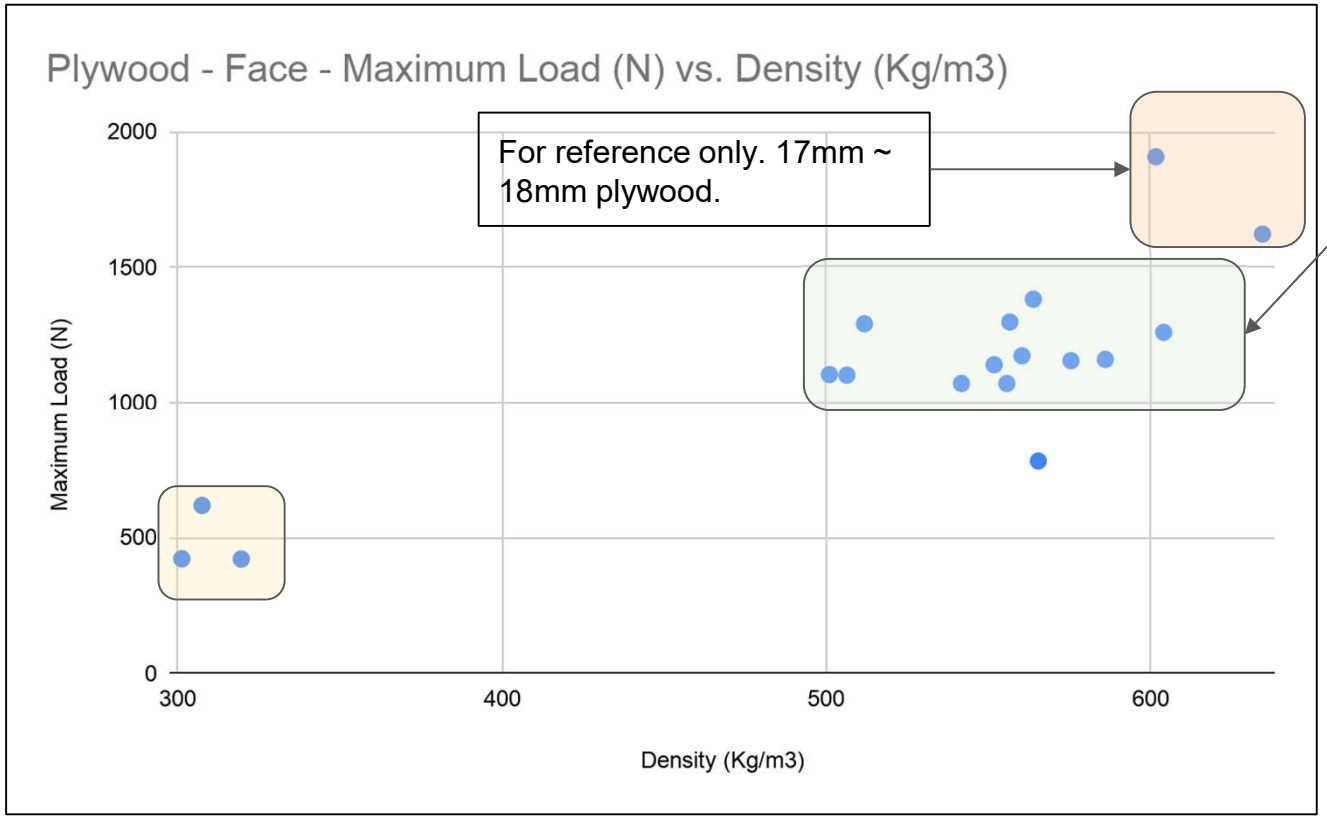
*fiber HK was hammermill 1 x with screen size 12mm/shieved to remove dust and flour dust/**Resin no heating 45C**/pmdi sg buloh/sc 95%/fiber length 2.82 cm(AVG)*

MOR range 35 ~ 40+
Sample:
*68, 69, *70 & *71*

**70 & 71 resin 11%.*
● *Resin 10% sample 56*
Resin no heating.

NOTE: ALL these sample data with TS at range 9% - 12%.

Oply Test Result - Analysis 3a - Plywood Screw Withdrawal Test



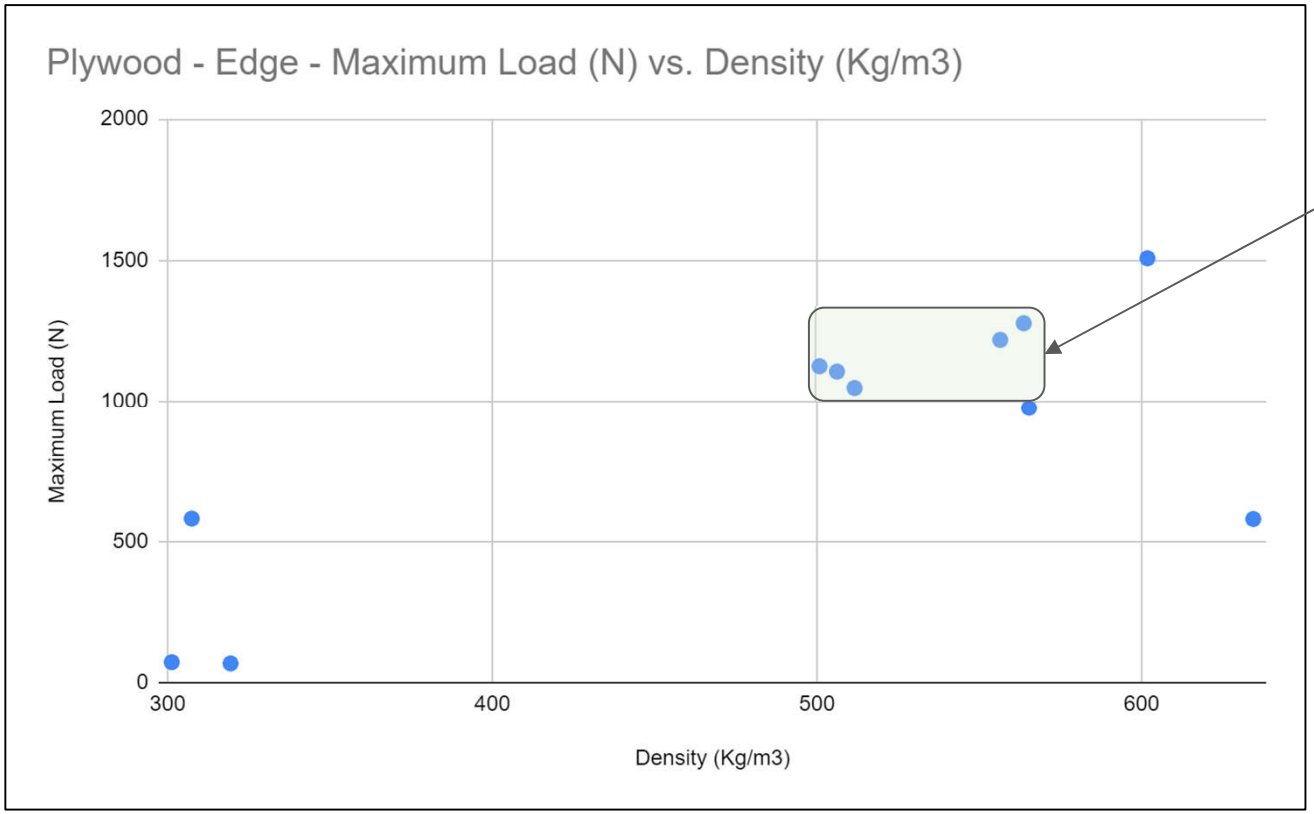
Plywood 11~12mm FACE screw withdrawal strength at the range of 1100N ~ 1400N.

(density at range of 500kg/m3 ~ 570kg/m3)

Plywood 17~18mm screw withdrawal strength at the range of 1100N ~ 1400N.

(density at range of 500kg/m3 ~ 570kg/m3)

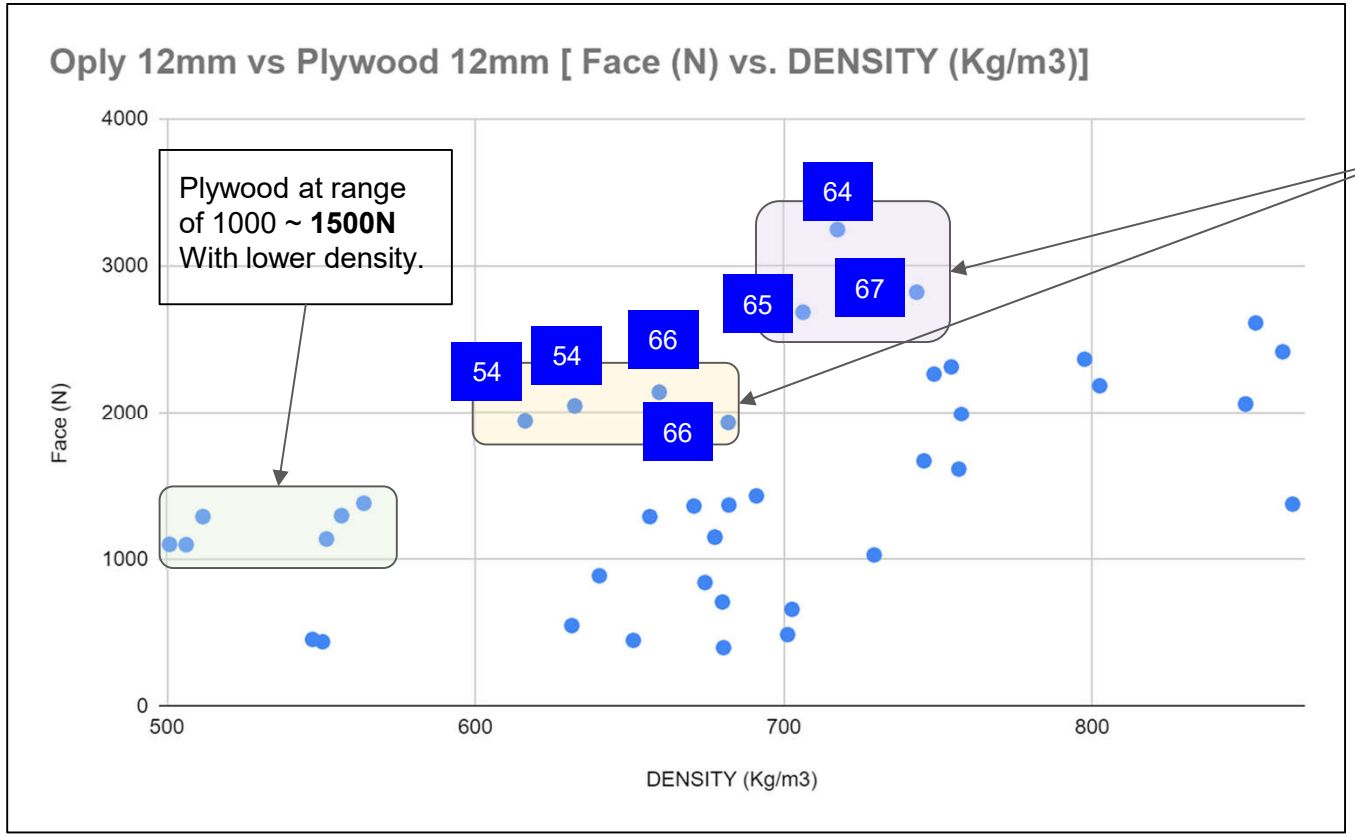
Oply Test Result - Analysis 3b - Plywood Screw Withdrawal Test



Plywood 11~12mm EDGE screw withdrawal strength at the range of 1100N ~ 1200N.

(density at range of 500kg/m3 ~ 570kg/m3)

Oply Test Result - Analysis 3c - Oply FACE Screw Withdrawal Test

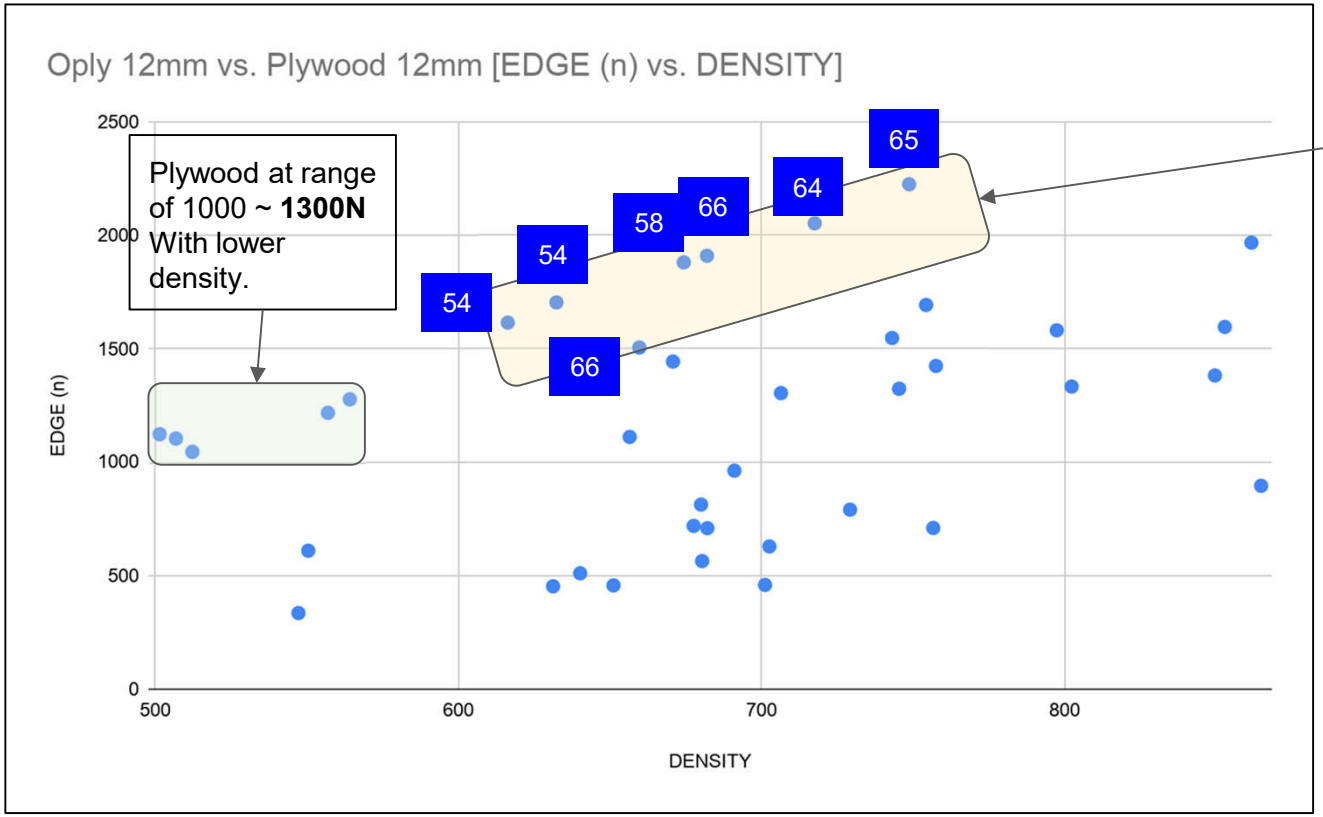


Both these range of show good result at 2000 ~ 3000N at higher density.

Potentially, we can optimize these two range of sample at lower density.

XX = sample number

Oply Test Result - Analysis 3d - Oply EDGE Screw Withdrawal Test



This range of show good result at 1500 ~ 2300N at higher density.

The load is proportional to density.